Autonomous Quadrotor Terrain-Following with a Laser Rangefinder and Gimbal System

Matthew Clark* and Robert C. Roberts†
*Corvidae Technology (HK) Limited, Hong Kong
†Department of Electrical and Electronic Engineering, The University of Hong Kong, Hong Kong
Email: matt@corvidaetechnology.com, robert.roberts@hku.hk

***Abstract*—This paper introduces a technique using a laser rangefinder sensor mounted on a movable gimbal system that enables an unmanned aerial vehicle (UAV) to maintain a specific height above changing terrain. The system and controller design are explained, and the performance of the laser rangefinder is compared with ultrasonic and GPS-based measurements.**

***Keywords*—LIDAR, quadcopter, rangefinder, UAV, gimbal, terrain-following, autonomous.**

I. INTRODUCTION

The ability to accurately measure an aircraft's height-above-ground (AGL) is necessary to ensure safe operation of that aircraft, especially while flying close to ground-level or during landing. For autonomous UAVs, knowledge of AGL helps improve safety as well as opens up new application possibilities.

Most current research into terrain-following algorithms for UAVs, such as presented in [1], [2], and [3], involves relying on a known elevation map of the surrounding terrain and having the UAV calculate optimal terrain-following routes before flight begins. The main disadvantage with these algorithms is that the UAV cannot react in real-time to the terrain changes. Other papers, such as [4], [5], and [6], have proposed vision-based terrain mapping procedures; however, vision techniques require the UAV to have a smooth flight to avoid motion blur and error in the terrain estimation increases with the UAV's height. Over the last few years, LIDAR systems and laser rangefinders have gotten more compact and lightweight, rendering them practical for use on a small UAV. [7] and [8] both propose using a spinning LIDAR system to create a full 3D terrain map, which is computationally intensive, and the point cloud calculations take place on a ground station computer instead of on the UAV itself. Silvagni, et al. in [9] propose using a laser altimeter with a UAV, but one disadvantage is that the laser is fixed on the bottom of the UAV, so as the UAV pitches forward during flight, the distances will be based on terrain behind the UAV.

This paper seeks to address the gaps in prior approaches by introducing a novel laser rangefinder and gimbal system to be used on a small quadcopter, which will allow or enhance the development of future applications relating to autonomous terrain-following.

II. HARDWARE SELECTION

A research-grade Matrice 100 quadcopter from Dà-Jiāng Innovations Science and Technology Co., Ltd (DJI) was selected along with DJI's Manifold embedded computer. The Manifold runs Linux and uses the Robot Operating System (ROS) framework for flight control, sensor interfacing and data telemetry transmission. During flight, the DJI Go smartphone application reports the quadcopter's height to the user. This "height" value is calculated by the flight controller on the Matrice 100, and is measured via GPS as the difference in height from the home point, the location from which the drone took off, so any change in ground elevation is not registered.

To physically detect the true AGL distance from the quadcopter, an SF11/C rangefinder (Lightware Optoelectronics, South Africa) is used. The SF11/C, can detect obstacles at up to 120m and weighs only 35g which is ideal for use on a quadcopter. Initially, the laser rangefinder was fixed pointing downward on the quadcopter.

To test the ground-detection ability in the field, the SF11/C was compared with DJI's provided "Guidance" sensor, which is their obstacle avoidance system composed of an ultrasonic sensor and stereo camera. Both sensors were mounted on the bottom of the Matrice 100, which was then flown vertically up to 25m and back down again above level ground. The distance readings for both sensors were plotted against DJI's height data given by the flight controller, as shown in Fig. 1. The Guidance sensor has detection problems after 5m and totally fails to detect the ground at 9 meters, while the SF11/C laser rangefinder readings are unbroken and very closely mimic the drone's actual height.

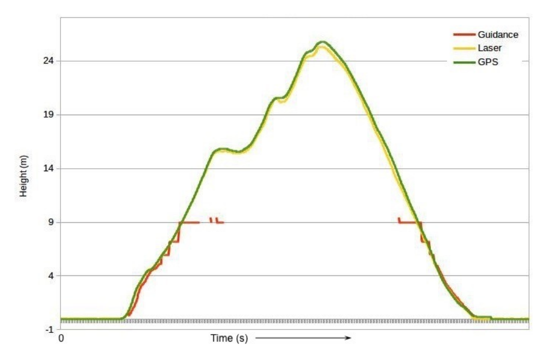

Fig. 1: Height sensor measurement comparison of the ultrasonic Guidance system(red), GPS (green), and laser rangefinder (yellow). Note errors above 5m, and distance limitation above 9m for the Guidance system.

One problem with a fixed sensor is that as the quadcopter moves forward, the body frame pitches forward and changes

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the angle at which the sensor detects the ground. To test this theory, the Matrice 100 was flown over a test area at a near-constant height of 2m above the ground. The laser distances were recorded and compared to DJI's given height data, and both measurements were plotted against pitch angle, as shown in Fig. 2. It is clear to see the laser distance is affected by the drone's pitch angle; when the pitch angle grows larger, the laser distance measurement increases.

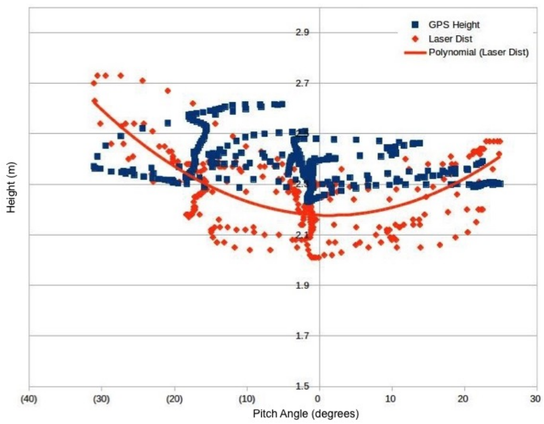

Fig. 2: SF11/C laser rangefinder data versus quadcopter pitch angle

III. Mounting the Rangefinder on a Gimbal

For ideal terrain detection, the sensor should point further in front of the drone the faster it moves, allowing more time to react to upcoming terrain as the drone moves quickly. To achieve this, the laser rangefinder is mounted on a movable gimbal so the pitch angle can be controlled according to the quadcopter's speed. As the laser rangefinder is approximately the same size and weight as a GoPro, a GH3-3D 3-Axis Camera Gimbal (Hobby King) is employed for sensor stabilization and control. A custom 3D-printed case was made to hold the rangefinder securely to the gimbal's GoPro slot, which ensures the rangefinder points straight down when the gimbal is level.

The gimbal was attached to the quadcopter via a carbon fiber plate. The sensor moves using a simple servo motor whose angle needs to be set according to the UAV's speed by a custom ROS node running on the Manifold, with an Arduino microcontroller used for gimbal interfacing. The gimbal mounted laser rangefinder system is shown schematically and mounted on the Matrice 100 in Fig. 3.

IV. Terrain-Following Algorithm

Assuming that the laser rangefinder and gimbal system gives accurate measurements of ground distance, the final step is implementing this information into the quadcopter's flight path to ensure adequate terrain-following. In ROS, a simple "movement framework" to control the movement of the quadcopter was designed, in which the drone receives a goal position in local x, y, z coordinates and moves toward that goal. A simple reactionary terrain-following algorithm can then be implemented, as follows:

1) Receive goal as $x_{goal}, y_{goal}, z_{goal}$ local coordinates

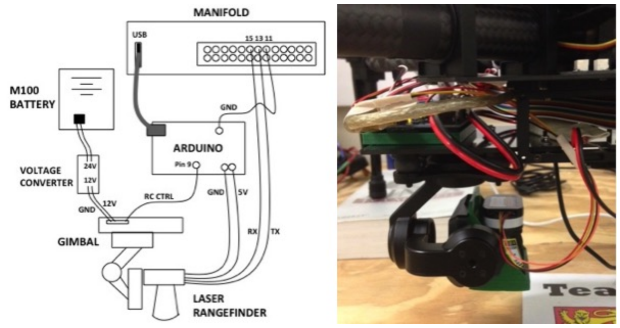

Fig. 3: Schematic of the gimbal and laser rangefinder system and final mounting on the Matrice 100

2) Set the AGL for the quadcopter (*tf_height*) equal to the z_{goal} coordinate
3) Take off and ascend to *tf_height*
4) Begin moving toward goal position
5) Detect a change in elevation (Δz) with laser rangefinder, determine x, y position of detection point
6) Calculate intermediate goal coordinates x_i, y_i, z_i of detection point where z_i is the new altitude that maintains *tf_height* at that point ($z_{current} + \Delta z$)
7) Move toward intermediate goal until x_i, y_i, z_i is reached, then continue moving toward final goal position. If another elevation change is detected, repeat steps 5-7.
8) Once position x_{goal}, y_{goal} is reached, stop.

This algorithm does not require intensive calculation so it can be performed in real-time on the Manifold of the Matrice 100 without using much computational power or battery. The intermediate goal positions are spaced according to the drone's forward speed to allow the UAV time to adjust to any change in elevation at those points. The user defines a vertical reaction time (t_{z_max}), and knowing the quadcopter's velocity (v_{xy}) and initial height (*tf_height*), the gimbal angle (θ) is calculated as follows:

$$\theta_{laser\_gimbal} = \arctan\left(\frac{v_{xy} \times t_{z\_max}}{tf\_height}\right) \quad (1)$$

This calculated *lasergimbal_angle* is published on a ROS topic, and a listener node receives this new angle and sends it over the serial port. The code running on the Arduino reads this angle and then moves the gimbal servo by the required amount. Given this angle, the laser distance reading as *laser_dist*, and assuming that the drone is moving forward toward the goal position, the change in height, Δz, is calculated as follows:

$$\Delta z = tf\_height - laser\_dist \times \sin(-\theta_{laser\_gimbal}) \quad (2)$$

Fig. 4 illustrates geometric relationships for the gimbal-mounted laser rangefinder. Point A on the ground represents the drone's current (x, y) position and point B is where the laser ray reflects from the ground. The ground distance between points A and B is represented by d_g:

$$d_g = laser\_dist \times \cos(\theta_{laser\_gimbal}) \quad (3)$$

Point B is the "intermediate goal" position for the drone, and x_B and y_B can be determined as follows, where ψ_{goal} is the yaw angle of the drone to the goal position:

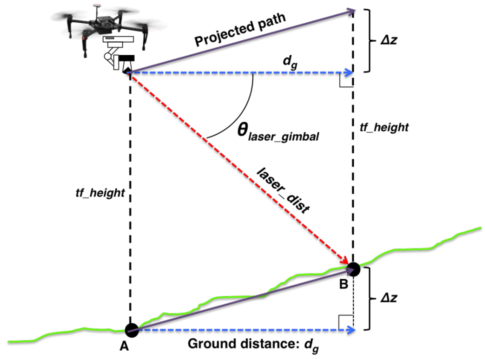

Fig. 4: Diagram of laser gimbal system showing geometric relations between components and the ground.

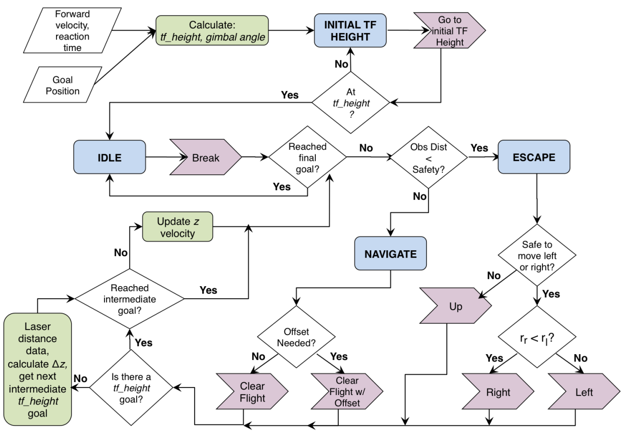

Fig. 5: State machine diagram describing the terrain-following algorithm. Blue rectangles represent mission states, pink arrows represent resulting drone action states, and green rectangles represent terrain-following processes.

$$x_B = d_g \times \cos(\psi_{goal}) + x_A \tag{4}$$

$$y_B = d_g \times \sin(\psi_{goal}) + y_A \tag{5}$$

The *z_goal* position is updated to reflect the new local z position by adding the Δz to the drone's current z position. The quadcopter continues toward the final goal position normally, maintaining the same x and y velocity components, but the vertical component of the velocity is updated as follows:

$$v_z = \frac{(z_{goal} - z_{current})}{max\_dist\_change\_z} \times v_{z\_max} \tag{6}$$

V. Results and Discussion

Most of the testing of the terrain-following algorithm was conducted in the DJI Assistant simulator. In the simulator, there is no terrain and the ground is completely flat. Therefore, to trick the simulator into thinking it was flying above changing terrain, the laser was pointed at different points on the wall, varying the distances. During the quadcopter's flight toward the goal position, its height above ground adjusted based on the varied laser distance readings, as expected. The gimbal on the physical drone also accurately moved to the correct angle. The gimbal motion is not as precise as the algorithm stipulates, and the actual gimbal angle varied by a few degrees. In the field, this potential error should be measured and then adjusted according to the user's preferences.

During testing in the lab and in the field, the laser rangefinder results were very consistent and accurate, and using the laser to determine height above ground is a simple and effective means to implement terrain-following. However, the one dimensionality of the laser means that small variations in the terrain may cause unwanted movements of the drone. Also, the laser is agnostic to what objects it detects: it will detect the tree canopy, people, and buildings, and those objects would be considered part of the "terrain". Finally, if the laser beam of the rangefinder hits a highly reflective surface, the distance readings may be inaccurate. Generally, the benefits and accuracy of the laser rangefinder outweigh these drawbacks, but it is good to be aware of and to compensate for them in future applications.

VI. Conclusion

A technique for precise terrain-following using a laser rangefinder is proposed. The laser rangefinder is mounted on a custom-designed gimbal, which is controlled by a separate arduino board, and the entire laser-and-gimbal system is attached on the underside of the Matrice 100. The gimbal allows the laser rangefinder to consistently point ahead of the UAV as it pitches forward during flight, allowing the terrain height to be measured ahead of the UAV's motion and giving the UAV time to adjust to any upcoming change in terrain. The algorithm is dependent on an accurate GPS signal to ensure precision of positioning, but overall the method proved successful and can be useful in certain terrain-following applications.

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